

Degree: B.Tech., Semester: Third-Course work
MID-SEMESTER EXAMINATION, September, 2024

Course Code: COMTC13

Course Title: Probability and Stochastic Processes

Duration: 1 hour 30 minutes

Max. Marks: 25

Note: Attempt all five questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO												
1(a)	A gambler plays a game of rolling a die with the following rules. He will win Rs. 200 if he throws a 6 but will lose Rs. 40 if he throws 4 or 5 and lose Rs. 20 if he throws 1, 2, or 3. Find the expected value that the gambler may gain.	2.5	CO1												
1(b)	Let X be a random variable taking values $-1, 0$ and 1 such that $P(X = -1) = 2P(X = 0) = P(X = 1).$ Find the mean of $Y = 2X - 5$.	2.5	CO1												
2(a)	The mileage C in thousands of miles which car owners get with a certain kind of tyre is a random variable having the PDF $f(x) = \begin{cases} \frac{1}{20} e^{-\frac{x}{20}} & x > 0 \\ 0 & x \leq 0. \end{cases}$ Find the probability that one of these tyres will last i) at most 10,000 miles, ii) at least 30,000 miles.	2.5	CO1												
2(b)	The pdf of a continuous random variable X is $f(x) = \begin{cases} a + bx^2 & 0 \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$ Find the constants a and b if $E(X) = \frac{3}{5}$.	2.5	CO1												
3(a)	The first four moments of a distribution about the value 4 are 1, 4, 10, and 45, respectively. Find the first four central moments.	2.5	CO1												
3(b)	Define Kurtosis. The first four moments of a distribution about the mean are 0, 2.5, 0.7, and 18.75. Find out the kurtosis and conclude the shape of the distribution curve.	2.5	CO1												
4(a)	Briefly explain the method of ordinates and the method of areas to fit a normal distribution for a given set of values.	2.5	CO2												
4(b)	Fit a Poisson's distribution to the following data and calculate theoretical frequencies: <table border="1" style="margin-left: auto; margin-right: auto;"><tr><td>Deaths</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Frequency</td><td>122</td><td>60</td><td>15</td><td>2</td><td>1</td></tr></table>	Deaths	0	1	2	3	4	Frequency	122	60	15	2	1	2.5	CO2
Deaths	0	1	2	3	4										
Frequency	122	60	15	2	1										
5(a)	Derive the mean and variance of Beta distribution of the first kind.	2.5	CO2												
5(b)	Does there exist a random variable X for which $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) = 0.6$, where μ and σ are the mean and standard deviation of X .	2.5	CO2												